

■ SignLens Mobile Application Subsystem

React Native (Expo SDK 54) Client Setup & User Guide

Welcome to the **SignLens Mobile Application Subsystem**. This is the cross-platform mobile client application built with **React Native (0.81.5)**, **Expo SDK 54**, **TypeScript**, **Expo Router**, **NativeWind (Tailwind CSS)**, and **Zustand**.

It provides real-time camera gesture recognition, text-to-speech vocalization, bidirectional voice/text-to-sign video playback, and a gamified Learning Management System (LMS).

■ 1. Directory Structure Overview

```
SignLens/frontend/
  ■■■ app/                                # Expo Router file-based screen routes
  ■ ■■■ (auth)/                          # Login, signup, forgot password, OTP screens
  ■ ■■■ (tabs)/                          # Main bottom tab bar navigation
  ■ ■ ■■■ camera.tsx                     # Real-time Sign Recognition Camera Scanner
  ■ ■ ■■■ home.tsx                       # Home Dashboard & Quick Action Launchers
  ■ ■ ■■■ learn.tsx                      # Gamified Sign Language Courses & Badges
  ■ ■ ■■■ settings.tsx                   # User profile & application settings
  ■ ■■■ translation-result.tsx           # Glassmorphic prediction sheet & Text-to-Speech (TTS)
  ■ ■■■ text-to-sign.tsx                 # Typed text to sign language video player
  ■ ■■■ voice-to-sign.tsx                # Speech-to-text to sign video player
  ■■■ components/                        # Reusable UI components (GlassCard, FormInput, Header)
  ■■■ hooks/                             # Custom React Hooks (usePredict, useAuth, useHistory)
  ■■■ services/                          # API HTTP client (apiClient.ts) & endpoint definitions
  ■■■ stores/                            # Zustand state management (authStore.ts)
  ■■■ package.json                       # Project dependencies & Expo scripts
```

■ 2. Step-by-Step Installation & Local Setup

Step 2.1: Prerequisites

- Ensure **Node.js (v18.0.0 or higher)** is installed on your computer.
 - Download the free **Expo Go** app from the **Google Play Store** (Android) or **Apple App Store** (iOS) if testing on a physical smartphone.
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Step 2.2: Environment Configuration (.env File Setup)

Navigate to the `frontend/` directory in Command Prompt / Terminal:

```
cd frontend
```

Create a new file named `.env` in the `frontend/` directory with the following content:

```
# Change IP Address below to match your PC's local network IP address
EXPO_PUBLIC_API_URL=http://192.168.1.100:8001/api/v1
EXPO_PUBLIC_WS_URL=ws://192.168.1.100:8001/api/v1
```

■ How to find your PC's local IP address:

- **On Windows:** Open Command Prompt, type `ipconfig`, and look for **IPv4 Address** (e.g., `192.168.1.100` or `10.0.0.15`).
 - **On macOS / Linux:** Open Terminal, type `ifconfig` or `ip a`, and look for `inet` under your active Wi-Fi interface.
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Step 2.3: Install Dependencies

In your terminal (inside the `frontend/` folder), run:

```
npm install
```

■ 3. Running the Mobile Application

Start the Expo Metro Bundler dev server:

```
npx expo start
```

Once executed, a **large QR Code** will appear in your terminal alongside a menu:

```
> Scan the QR code above with Expo Go (Android) or the Camera app (iOS)

> Press a ■ open Android
> Press i ■ open iOS simulator
> Press w ■ open web browser
> Press r ■ reload app
```

Options to Test the App:

■ Option A: On a Physical Smartphone (Recommended)

- Ensure your smartphone and host computer are connected to the **same Wi-Fi network**.
- Open **Expo Go** on Android (or native Camera app on iPhone) and scan the terminal QR Code.
- The SignLens mobile app will load instantly on your phone!

■ Option B: In Web Browser

- In the terminal running `npx expo start`, press `w`.
- Google Chrome will automatically launch `http://localhost:8081` with the app UI layout.

■ Option C: On Android Studio Emulator

- Launch your Android Emulator in Android Studio.
- In the terminal running `npx expo start`, press `a`.

■ 4. Key Mobile App Features Walkthrough

- **Camera Real-Time Scanner** (`app/(tabs)/camera.tsx`):
 - Tap the Camera icon on the bottom navigation bar.
 - Point the camera at a signer performing a sign.
 - The app captures gesture frames and streams them to the server.
 - A glassmorphic result modal slides up showing the prediction (e.g., "THANK YOU") with confidence score.
 - Tap the **Speaker icon** to hear the translation spoken aloud via **Text-to-Speech** (`expo-speech`).
- **Text & Voice to Sign** (`app/text-to-sign.tsx` & `app/voice-to-sign.tsx`):
 - Type text (e.g., "Hello") or speak into the microphone.
 - The app queries the backend and streams sequential sign language videos rendered via `expo-video`.
- **Gamified Learning Hub** (`app/(tabs)/learn.tsx`):
 - Browse categories (Alphabets, Greetings, Numbers).
 - Complete interactive lessons, take flashcard quizzes, and earn achievement badges.

■ 5. Common Mobile Issues & Troubleshooting

■ Problem 1: App displays "Network Error" when translating signs

- **Cause:** The mobile app is trying to connect to `localhost`, which refers to the phone itself rather than your PC server.
- **Solution:**
 - Find your PC's IP address (`ipconfig` on Windows).
 - Update `frontend/.env` with your actual IP address (e.g., `EXPO_PUBLIC_API_URL=http://192.168.1.100:8001/api/v1`).
 - Clear Expo cache and restart: `npx expo start -c`.

■ Problem 2: Camera permissions denied or black screen

- **Cause:** Camera permission was not granted on smartphone.
- **Solution:** Go to Phone Settings $\rightarrow$ Apps $\rightarrow$ Expo Go $\rightarrow$ Permissions $\rightarrow$ Enable Camera permission.

■ Problem 3: Metro Bundler fails to start (Unable to resolve module)

- **Cause:** Stale `node_modules` or Expo cache.
- **Solution:** Run `npx expo start -c` to reset the Metro bundler cache.